GLUEING DATA FOR SHEAVES OF ABELIAN GROUPS, ALGEBRAIC STRUCTURES OR MODULES PRODUCES A SHEAF IN THE RELEVANT CATEGORY

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Allowed standard prerequisites (editorial).

- The set-level glueing construction of lemma-glue-sheaves, whose sections over W are the equalizer of the two maps prod F_i(W cap U_i) -> prod F_i(W cap U_i cap U_j). invoked at proof lines 5483-5494. Fixes the set-level description that gets enriched with structure.

Source reference. Original lemma 00AL: sources/stacks-sheaves/pinned-original-source.tex, source lines 5409–5409. Cited at statement lines 5476-5478 and used throughout the proof for the set-level construction; load-bearing inside the counted span.

Lemma 0.2. *Let X be a topological space. Let $X = \bigcup U_i$ be an open covering. Let $(\mathcal{F}_i, \varphi_{ij})$ be a glueing data of sheaves of abelian groups, resp. sheaves of algebraic structures, resp. sheaves of $\mathcal{O}$-modules for some sheaf of rings $\mathcal{O}$ on X. Then the construction in the proof of Lemma 00AL above leads to a sheaf of abelian groups, resp. sheaf of algebraic structures, resp. sheaf of $\mathcal{O}$-modules.*

Proof. This is true because in the construction the set of sections $\mathcal{F}(W)$ over an open W is given as the equalizer of the maps

$$\prod_{i \in I} \mathcal{F}_i(W \cap U_i) \rightrightarrows \prod_{i,j \in I} \mathcal{F}_i(W \cap U_i \cap U_j)$$

And in each of the cases envisioned this equalizer gives an object in the relevant category whose underlying set is the object considered in the cited lemma. □

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